REPORT FOR LAB1

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Performance Report: Interactive Image Mosaic Generator

Method

The mosaic generator reconstructs an input image by:

1. **Preprocessing**
    - Input is cropped so width/height are multiples of the chosen **grid size** (16, 32, 64).
    - No quantization was applied in these experiments.
2. **Grid segmentation**
    - Each cell's mean LAB color is computed.
    - Both **vectorized NumPy** and **loop-based** implementations are executed.
3. **Tile preparation**
    - 4096 CIFAR-100 tiles (32×32) were used.
    - Each tile's mean LAB color is precomputed.
4. **Tile mapping**
    - Each cell is matched to the nearest tile in LAB space.
    - Vectorized vs. loop runtimes are compared.
5. **Mosaic output & metrics**
    - Final mosaics are assembled.
    - Metrics: **MSE** (pixel-wise error) and **SSIM** (structural similarity).

Results (Similarity + Runtime)

Grid Size	Tiles Used	MSE	SSIM	Vectorized Time (s)	Loop Time (s)
16×16	4096	2205.15	0.1089	0.007	0.028
32×32	4096	2005.02	0.0823	0.083	0.112
64×64	4096	1796.81	0.0698	0.138	0.505

Performance Sweep (Vectorized vs. Loop)

Grid Size	Vectorized Time (s)	Loop Time (s)	Speedup
16×16	0.007	0.028	~4×
32×32	0.083	0.112	~1.3×
64×64	0.138	0.505	~3.6×

Observations

- **Vectorization advantage**
  - 2–4× faster across all grid sizes.
  - Critical at larger grids (64×64: 0.138s vs. 0.505s).
- **Grid size trade-off**
  - **16×16**: Highest SSIM (0.1089), but slightly higher MSE.
  - **32×32**: Balanced runtime (~0.08s vectorized) and SSIM (0.0823).
  - **64×64**: Lowest MSE (1796.81) but SSIM drops to 0.0698 (more blocky).
- **Metric differences**
  - **MSE decreases** with larger grids (better pixel-wise fit).
  - **SSIM decreases** with larger grids (less structural similarity).
  - SSIM aligns better with perceived quality in mosaic tasks.

Conclusion

- **Vectorization is essential** for interactivity, especially at higher grid sizes.
- **Loop implementation** serves as a correct reference but is inefficient.
- **Best balance**: **32×32 grid** — moderate runtime, reasonable SSIM, acceptable visual detail.
- **For evaluation**: SSIM should be prioritized over MSE, as it better captures perceptual similarity.